Experimental Plan

Usability Evaluation of an Urdu LLM-Assisted Touch Interface for a Simulated Smart Home

Group member	Student ID
Muhammad Ibrahim	29079
Ashhal Aamir	29114
Faraz Ansari	29201

Purpose and objectives

This small exploratory study will evaluate whether participants can understand and use a bilingual Urdu/English touch interface to complete common smart-home tasks. The prototype uses a language model to offer Urdu suggestions and a local simulator to represent device changes. The study focuses on usability, clarity of wording, task completion, and participant feedback; it does not evaluate real smart-home hardware.

Research questions

1. How successfully can participants complete the five assigned smart-home tasks using the interface? 2. How do participants describe the clarity and usefulness of its Urdu/English labels and suggestions? 3. What usability, accessibility, or privacy improvements do participants recommend?

Study design and audience

A single-condition, moderated usability study will be conducted with a small convenience sample of approximately 5-8 adults from the intended user audience: people who are Deaf or Hard of Hearing and use Urdu, English, Pakistan Sign Language, or a mix of communication modes. The sample is intended for early feedback and feasibility, not statistical generalization. Participants may use their preferred communication mode and reasonable accessibility accommodations.

Prototype and tasks

Participants will use the running Gradio application on a computer or tablet. It presents large bilingual touch choices, Groq-generated Urdu action/suggestion options when available, and a simulated home-state panel. If the service is unavailable or produces unusable output, the interface uses a fixed Urdu fallback. No physical devices are controlled.

Task	Assigned goal
1	Check today's weather in Lahore.
2	Set a five-minute timer for tea.
3	Set the light to blue.
4	Show a biryani recipe.
5	Set a reminder for Fajr prayer.

Session procedure

One participant at a time | approximately 30-40 minutes

Step	What happens	Time
1. Welcome and consent	Explain the study, voluntary participation, privacy, and the right to stop or skip questions. Obtain consent before entering study data.	4 min
2. Background and setup	Assign a participant code, collect brief background information, check accessibility preferences, and explain that the home is simulated.	4 min
3. Exploration	Allow a short, unassisted look at the interface. Give neutral help only if needed.	5 min
4. Five tasks	Present the five goals in the same order for each participant. Observe independently; record completion, help, errors, and task time. The participant confirms the simulated command or marks the task incomplete.	10-15 min
5. Questionnaire	Administer the 10-item SUS plus the adjective rating and NPS items. Use the provided bilingual materials and avoid leading responses.	4-5 min
6. Interview and close	Ask about the easiest/hardest part, Urdu wording, AI/privacy concerns, hands-free alternatives, and suggested improvements. Thank the participant and save the session.	5-7 min

Measures and analysis

Quantitative: task completion status and latency; interaction choices and model/fallback status; 10 SUS responses and calculated SUS score; adjective rating (1-7); and NPS (0-10). Report participant counts, completion percentages, and descriptive summaries such as median/mean task time and SUS. With a small sample, results will be treated as descriptive and exploratory.

Qualitative: summarize interview and researcher notes using a small set of themes, such as wording clarity, navigation, accessibility, trust/privacy, and desired improvements. Use short anonymized quotations only when permission and the study materials allow it.

Data handling and ethics

Use a participant code rather than a name in the app. Session records are stored locally as JSON and CSV files for analysis. Keep the files in a restricted project folder and do not publish identifiable details. Participation is voluntary; participants may stop or skip questions. Any audio/video recording is optional, requires separate consent, and is managed separately because the app does not record audio.

Scope and limitations

This is an early pilot of one touch-interface condition with a small sample and five simulated tasks. It has no comparison group, no real connected devices, and no voice or sign-language recognition. Groq output and network latency can vary; fallback use will be recorded. Findings will guide prototype improvements and should not be presented as broadly generalizable evidence.

Expected Phase 1 output: a running prototype and an initial, transparent usability plan for collecting and analyzing small-sample feedback.